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**SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE**

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

**AUTONOMOUS EXAMINATIONS - MAY'2025**

**COMMON TO B.E – ECE / CSE / B.TECH - IT & M.TECH – CSE**

**20EC211 - EMBEDDED SYSTEMS AND INTERNET OF THINGS**

**Answer ALL questions**

**Duration: 3 Hours**

**Maximum: 100 Marks**

**PART - A**

**(7x 1 = 7 Marks)**

1. **CO1 What is the role of an emulator in embedded systems?**
  - a) Physically replace the microcontroller
  - b) Copy software
  - c) Format memory
  - d) Run high-level software
2. **CO1 Which component is NOT typically an external peripheral in embedded systems?**
  - a) Arithmetic Logic Unit (ALU)
  - b) Flash memory
  - c) LCD screen
  - d) Temperature sensor
3. **CO2 Which of the following is a typical constraint in embedded systems?**
  - a) Unlimited power supply
  - b) High processing speed with no time constraints
  - c) Real-time performance
  - d) No memory restrictions
4. **CO2 Which of the following is a popular Embedded Linux distribution?**
  - a) Fedora
  - b) Ubuntu
  - c) Yocto
  - d) Debian
5. **CO2 In the architecture of Embedded Linux, which part interacts directly with the hardware?**
  - a) Shell
  - b) Kernel
  - c) User Space
  - d) Application Layer

6. **CO3** What is the correct order of the Linux startup sequence in an embedded system?
- a) Application → Bootloader → Kernel → Init
  - b) Kernel → Bootloader → Init → Application
  - c) Init → Bootloader → Kernel → Application
  - d) Bootloader → Kernel → Init → Application
7. **CO4** The scheduler in an RTOS is responsible for \_\_\_\_\_.  
 a) Task management and execution order      b) Memory management  
 c) File system management                      d) Device handling

**Fill in the blanks:**

**(3x 1 = 3 Marks)**

8. **CO4** The \_\_\_\_\_ is responsible for determining which task will run next in a real-time operating system.
9. **CO5** The essential characteristic of cloud computing that allows users to scale their computing resources based on demand is \_\_\_\_\_.
10. **CO6** The motion detector sends alerts to the \_\_\_\_\_ which processes the information and takes necessary actions.

**PART - B**

**(5x 3 = 15 Marks)**

11. **CO1** Define compiling, linking, and locating.
12. **CO3** What are the advantages of using Embedded Linux?
13. **CO3** Describe the Linux startup sequence in brief.
14. **CO4** What is Priority Inversion and how is it avoided?
15. **CO5** Name any three components involved in the physical design of an IoT system.

**PART - C**

**(5 x 15 = 75 Marks)**

**Q.No: 16 Compulsory Question**

**Q.No: 17 to 22 Answer any FOUR Questions**

**Compulsory Question:**

16. **CO3** i) Describe the components and role of a GNU cross-platform toolchain. (8)
- CO3** ii) What are device drivers? Explain different types of device drivers in Linux. (7)

17. **CO2** i) Explain memory testing and programming of flash memory in embedded systems. (8)
- CO2** ii) Outline the MISRA C guidelines and their importance in embedded systems. (7)
18. **CO1** i) Explain the role of the infinite loop in embedded programming with an example. (8)
- CO1** ii) Discuss downloading and debugging methods in embedded systems. (7)
19. **CO4** i) Explain the concept of Semaphores. How are they used for mutual exclusion give an example? (8)
- CO4** ii) What is context switching in real-time systems? What are its overheads? (7)
20. **CO4** i) Explain different methods of inter-task communication in real-time systems. (8)
- CO4** ii) Explain the difference between Preemptive and Non-Preemptive Kernels with examples. (7)
21. **CO5** i) What is the Internet of Things (IoT)? Explain its key components and applications. (8)
- CO5** ii) Explain M2M (Machine-to-Machine) communication and how it relates to IoT. (7)
22. **CO6** i) Explain the working of an Air Quality Monitoring System (AQMS) in IoT. (8)
- CO6** ii) Explain the concept of Landslide Detection and Disaster Management using IoT. (7)

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